

Kubeadm to Kubespray Migration

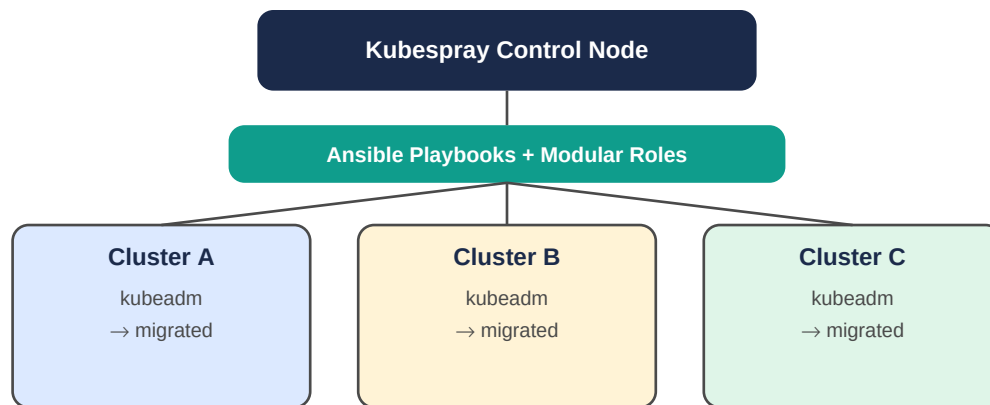
DevOps & Cloud Engineering Case Study

70% Less Configuration Effort	100% Automated, Repeatable Deploys	Multi Cluster-Ready Modular Roles
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Overview

Migrated an existing kubeadm-based Kubernetes cluster to Kubespray, improving cluster reliability, scalability, and lifecycle management. Manual shell-script workflows were replaced with Ansible-based automation, standardizing how clusters are provisioned and maintained.

Architecture



Key Deliverables

- Migrated the existing kubeadm-based Kubernetes cluster to Kubespray, improving cluster reliability, scalability, and long-term lifecycle management.
- Replaced manual shell-script workflows with Ansible-based automation, reducing configuration effort by 70% and minimizing human error during provisioning and upgrades.
- Modularized the entire deployment codebase into reusable Ansible roles, making cluster provisioning consistent and easier to manage across multiple clusters.
- Standardized cluster lifecycle operations (scaling, node add/remove, upgrades) through version-controlled, repeatable playbooks instead of ad-hoc scripts.

Outcome

A standardized, version-controlled Kubernetes deployment pipeline that replaces fragile, manual cluster setup with reusable Ansible automation — reducing configuration effort, minimizing human error, and making multi-cluster management consistent and repeatable.

Skills & Tools

Kubernetes · Kubespray · Ansible · kubeadm · Infrastructure as Code (IaC) · Configuration Management · Linux · Cluster Lifecycle Management · Shell Scripting · CI/CD